SCIENTIFIC JOURNAL PUBLICATION TREND TRACKING SYSTEM

Group 6

I. INTRODUCTION

1 Overview

1.1 Project Information

- Project Name: Scientific Journal Publication Trend Tracking System
- Group Name: Group 6
- Software Type: Web Application
- GitHub Repository:
  `https://github.com/NguyenNgocBaoTram10/Scientific-Journal-Publication-Trend-Tracking-System`

1.2 Project team

Full Name	Role	Student ID	Email
Lê Thanh Tú	Leader	049206004778	tult4778@ut.edu.vn
Nguyễn Ngọc Bảo Trâm	Member	082306008015	tramnnb8015@ut.edu.vn
Lê Thị Huyền Trân	Member	077306003635	tranlth3635@ut.edu.vn
Trần Trung Hiếu	Member	089206008473	hieutt8473@ut.edu.vn
Trần Văn Học	Member	052206005618	hoctv5618@ut.edu.vn
Nguyễn Đoàn Nhật Quang	Member	079206010585	quangndn0585@ut.edu.vn

Bảng 1: Project Team Information

2 Project Scope and Limitations

2.1 Major Features

2.1.1 Major Features for User

- Register and log in to the system.
- Search scientific publications using keywords, authors, journals, or research fields.
- View detailed publication information.
- Save publications to bookmarks.

- Follow research topics of interest.
- Receive notifications about new publications and research trends.
- View publication trend analysis and statistical dashboards.
- Generate and export analytical reports (PDF/CSV).

2.1.2 Major Features for Administrator

- Log in and log out.
- Manage user accounts.
- Configure academic API sources (Semantic Scholar, OpenAlex, and Crossref).
- Synchronize publication data from external APIs.
- Monitor the data synchronization process.
- Manage system notifications.
- Generate system reports.
- Configure system settings.

2.2 Project Limitations

2.2.1 Limited Use of Full-Text Data

- The system only uses publication metadata such as title, abstract, keywords, authors, and publication year.
- It does not process full-text content, which limits the depth of analysis and understanding of research papers.

2.2.2 Restricted Data Scope

- The system focuses only on selected research fields (e.g., Computer Science and Artificial Intelligence).
- Therefore, it does not fully represent trends across all scientific disciplines.

2.2.3 Dependence on External APIs

- The system relies on external APIs such as Semantic Scholar, OpenAlex, and Crossref.
- Any API downtime, structural changes, or request limitations may affect system stability and data collection performance.

2.2.4 Simplified Trend Analysis and Update Frequency

- Trend analysis is mainly based on publication frequency and growth rate.
- It does not apply advanced Machine Learning or AI models for prediction.
- Data is synchronized periodically (daily or weekly), so real-time trends may not be immediately reflected.

II. PROJECT MANAGEMENT PLAN

1 Overview

1.1 Scope and estimation

#	WBS Item	Complexity	Duration (Days)
1. Project Initiation			
1.1	Define project scope	Medium	1
1.2	Collect and analyze requirements	Complex	2
2. Project Management and Environment Setup			
2.1	Set up development environment (GitHub, Database, Backend, Frontend)	Simple	1
2.2	Design database (ERD, Database Schema)	Complex	2
2.3	Configure security (JWT, RBAC, Password Hashing)	Complex	2
3. System Design			
3.1	Design Use Case Diagram and Activity Diagram	Medium	1
3.2	Design Sequence Diagram	Medium	1
3.3	Design DFD and ERD	Medium	1
4. System Development			
4.1	User Registration and Login	Medium	2
4.2	Search Scientific Publications	Complex	2
4.3	View Publication Details	Medium	1
4.4	Bookmark Management	Medium	1
4.5	Research Topic Following	Medium	1
4.6	Notification Management	Medium	1
4.7	Publication Trend Analysis	Complex	3
4.8	Develop Statistical Dashboard	Complex	2
4.9	Display Trending Topics	Medium	1
4.10	Generate Statistical Reports (PDF/CSV)	Complex	2

#	WBS Item	Complexity	Duration (Days)
5. Administration Functions			
5.1	User Management	Complex	2
5.2	Configure Academic API Sources	Complex	2
5.3	Synchronize Data from Semantic Scholar, OpenAlex, and Crossref	Complex	3
6. Testing			
6.1	Prepare Test Plan	Medium	1
6.2	Design Test Cases	Medium	2
6.3	Execute Testing	Complex	2
6.4	Fix Defects After Testing	Complex	2
7. Project Finalization			
7.1	Complete Project Documentation	Medium	2
7.2	Deploy the System	Medium	1
7.3	Prepare Presentation Slides	Simple	1
		Total Duration	**46 Working Days**

1.2 Project Objective

- **Timeline:** The project must be completed before **July 3, 2026**.
- **Allocated Effort:** 46 man-days.

1.3 Project Risks

#	Risk Description	Impact	Possibility	Response Plan
1	External academic APIs (Semantic Scholar, OpenAlex, Crossref) become unavailable or change their endpoints.	The system cannot retrieve publication data, causing incomplete search results and inaccurate trend analysis.	High	Implement API fallback mechanisms, monitor API status regularly, and cache previously retrieved data whenever possible.

#	Risk Description	Impact	Possibility	Response Plan
2	Large volumes of publication data lead to slow system performance.	Users experience long response times when searching publications or viewing dashboards.	Medium	Optimize database indexing, implement pagination, caching, and asynchronous background synchronization.
3	Security vulnerabilities expose user accounts or stored data.	Unauthorized access, data leakage, and loss of user trust.	Medium	Use JWT authentication, RBAC authorization, password hashing, HTTPS, and regular security testing.
4	Trend analysis results become inaccurate because of outdated or incomplete datasets.	Generated reports and statistics may mislead users' research decisions.	Medium	Schedule automatic data synchronization and validate collected data before analysis.
5	System integration between frontend, backend, and database encounters compatibility issues.	Development delays and unexpected system failures during deployment.	Medium	Perform integration testing continuously and maintain version compatibility among all components.
6	Insufficient testing before deployment leaves critical defects unresolved.	System crashes, incorrect functionality, and poor user experience.	High	Prepare comprehensive Test Plans, Test Cases, conduct functional, integration, and user acceptance testing before release.
7	Project schedule delays due to underestimated implementation effort.	Failure to complete the project within the planned timeline.	Medium	Monitor project progress weekly, prioritize critical features, and adjust the implementation schedule when necessary.

2 Project Deliverables

Sprint	Week	Duration	Deliverables	Note
Sprint 1	Week 1	14/05 – 20/05/2026	• Define project scope • Analysis requirement	
Sprint 2	Week 2	21/05 – 27/05/2026	• Finish Product Overview • Finish User Requirement • Finish Functional Requirement	
Sprint 3	Week 3	28/05 – 03/06/2026	• Finish Database Design • Finish Software design description	
Sprint 4	Week 4	04/06 – 10/06/2026	• Finish Software requirement specification	
Sprint 5	Week 5	11/06 – 17/06/2026	• Finish Software design description (continued)	
Sprint 6	Week 6	18/06 – 24/06/2026	• Finish Software testing document • Finish Release package and user guide	
Sprint 7	Week 7	25/06 – 01/07/2026	• Testing • Finish document • Submitting	

3. **Responsibility Assignment**

Do~D; Review~R; Support~S

Responsibility	Thanh Tú	Bảo Trâm	Huyền Trân	Trung Hiếu	Văn Học	Nhật Quang
Project Planning & Tracking	S	D	S	S	S	S
Prepare Project Introduction Document	R	D	R	R	R	R
Prepare Software Requirement Document	S	D	S	S	S	S
Prepare Software Design Document	D	D	D	D	D	D
Prepare Test Document	D	D	D	D	D	D
Prepare Software User Guides	D	D	D	D	D	D
Prepare Final Report	D	D	D	D	D	D

4. **Project Communications**

Communication Item	Who	Purpose	When	Type
Team Weekly Meeting	Group	• Report work project • Assign work for next week • Control project deadline	Every Sunday	Meeting on Google Meet
Weekly Report	Member	• Report work process	1 day / week	Report via Zalo
Group Meeting	Group	• Report work process	Every Friday	Offline on class

5. Configuration Management

5.1 Document Management

- Document tool: Confluence
- Task management tool: Jira

5.2 Source Code Management

- Source code management tool: GitHub

5.3 Tools & Infrastructure

Category	Tools / Infrastructure
Programming Languages	PHP, HTML5, CSS3, JavaScript
Database	MySQL
Database Administration	phpMyAdmin
Academic APIs	Semantic Scholar API, OpenAlex API, Crossref REST API
API Testing	Postman
Version Control	Git, GitHub
Project Management	Jira, Confluence
Diagramming	Draw.io, PlantUML
IDE / Code Editor	Visual Studio Code
Local Server	WAMP Server

III. SOFTWARE REQUIREMENT SPECIFICATION

1 Product Overview

The Scientific Journal Publication Trend Tracking System (SJPTTS) is a web-based application developed to help researchers, students, lecturers, and academic institutions search, monitor, and analyze scientific publication trends. The system integrates academic data from Semantic Scholar, OpenAlex, and Crossref to provide comprehensive and up-to-date publication information.

The system allows users to search scientific publications using keywords, authors, journals, research fields, or publication years. Users can view detailed publication information, bookmark publications, follow research topics, receive notifications about newly published papers, analyze publication trends through interactive dashboards, and export reports in PDF or CSV format.

Administrators can manage user accounts, configure academic API sources, synchronize publication data, monitor system operations, and maintain the overall performance of the platform. The system aims to provide an efficient solution for discovering scientific literature and supporting research through publication trend analysis.

2 User Requirements

2.1 Actors

2.2 Use Cases

2.2.1 Diagram(s)

2.2.2 Descriptions

2.3 Functional Requirements

2.3.1 User Authentication and Authorization

- Users can register a new account.
- Users can log in using a valid username and password.
- The system assigns permissions based on user roles.
- Administrators can manage user accounts and roles.

2.3.2 Research Paper Search

- Users can search research papers by keyword, author name, or journal name.

- The system displays matching publication results.
- Users can view detailed publication information.

2.3.3 Bookmark Management

- Users can save research papers to personal bookmarks.
- The system prevents duplicate bookmarks.
- Users can access bookmarked papers later.

2.3.4 Publication Trend Analysis

- Users can analyze publication trends based on keywords, topics, authors, or journals.
- The system generates trend statistics from publication metadata.
- The system visualizes trends using charts and dashboards.

2.3.5 Dashboard Visualization

- Users can view publication statistics through dashboards.
- The dashboard displays publication counts, topic distributions, journal distributions, and growth trends.

2.3.6 Trending Topics

- Users can view trending research topics.
- The system ranks topics according to publication frequency and growth rate.

2.3.7 Topic Following

- Users can follow research topics of interest.
- The system stores followed topics for each user.
- Users may follow multiple topics.

2.3.8 Notification Management

- Users receive notifications related to followed topics.
- Notifications include new publications, trend changes, and emerging topics.
- Users can view notification details.

2.3.9 Report Generation

- Users can generate analytical reports based on publication trends.
- Reports summarize publication statistics and research trends.
- Reports are generated from publication metadata.

2.3.10 System Administration

- Administrators can manage system operations.
- Administrators can manage user accounts.
- Administrators can configure academic API sources.
- Administrators can synchronize publication metadata from external APIs.

2.3.11 Academic Data Synchronization

- The system synchronizes publication metadata from external academic APIs.
- The system validates retrieved metadata.
- The system inserts new records and updates existing records.
- The system records synchronization logs and statistics.

2.4 Non-Functional Requirements

2.4.1 Usability

The interface is intuitive and accessible for all user types.

2.4.2 Performance

Search results must be returned within 2 seconds.
The system must support at least 500 concurrent users.

2.4.3 Reliability

System uptime must be at least 99% during business hours.
Data must be backed up daily.

2.4.4 Security

User passwords must be hashed before storage.
Role-based access control must be enforced for all features.

2.4.5 Maintainability & Scalability

The system must use a modular architecture for easy maintenance.
New data sources can be added without redesigning the existing system.

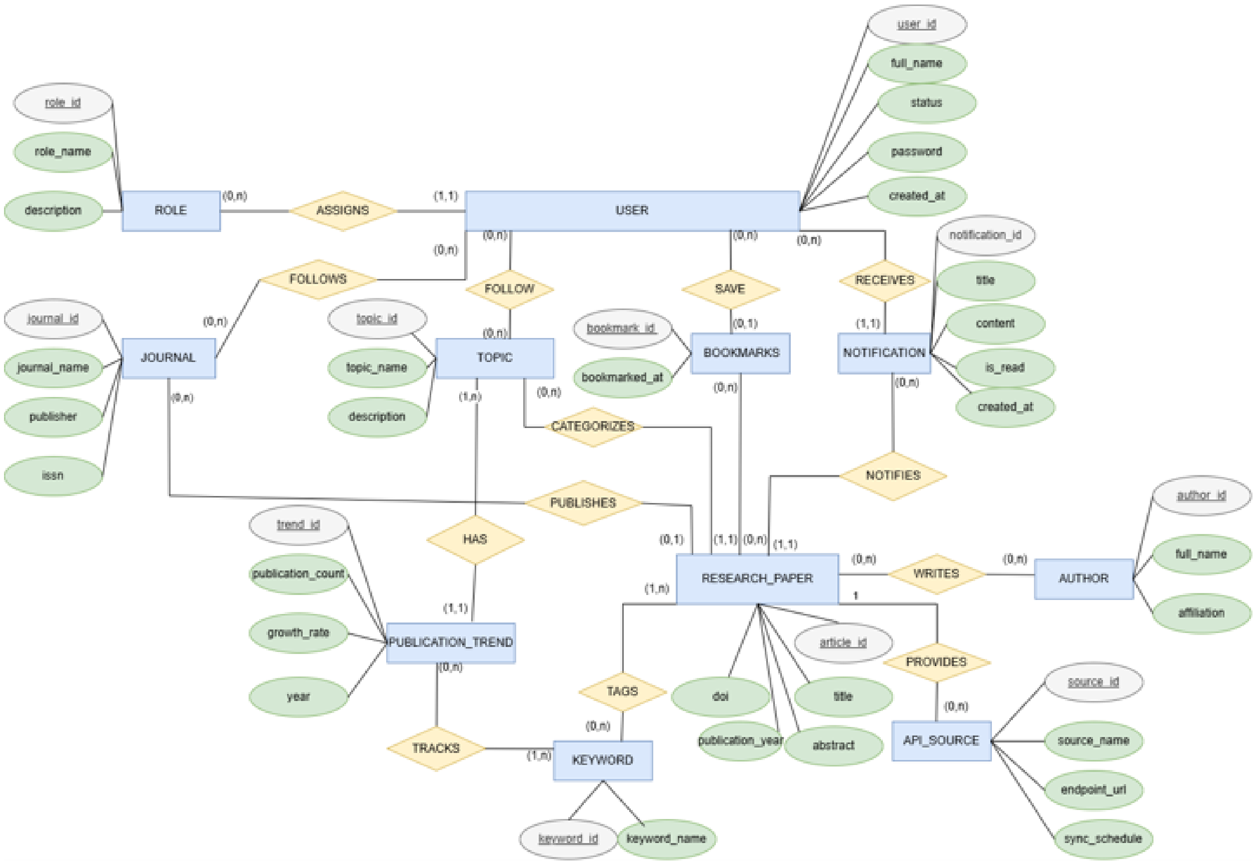
2.5 Screen Authorization

Screen	Researcher	Lecturer/Stude	System Administrator
Login	x	x	x
Register	x	x	
Search Research Papers	x	x	
View Paper Details	x	x	
Save Bookmarks	x	x	
Track Publication Trends	x	x	
View Dashboard	x	x	
Generate Reports	x	x	
View Trending Topics	x	x	
Follow Topics	x	x	
Receive Notifications	x	x	
Manage System			x
Manage Users			x
Configure API Sources			x
Sync Academic Data			x

2.6 Non-Screen Functions

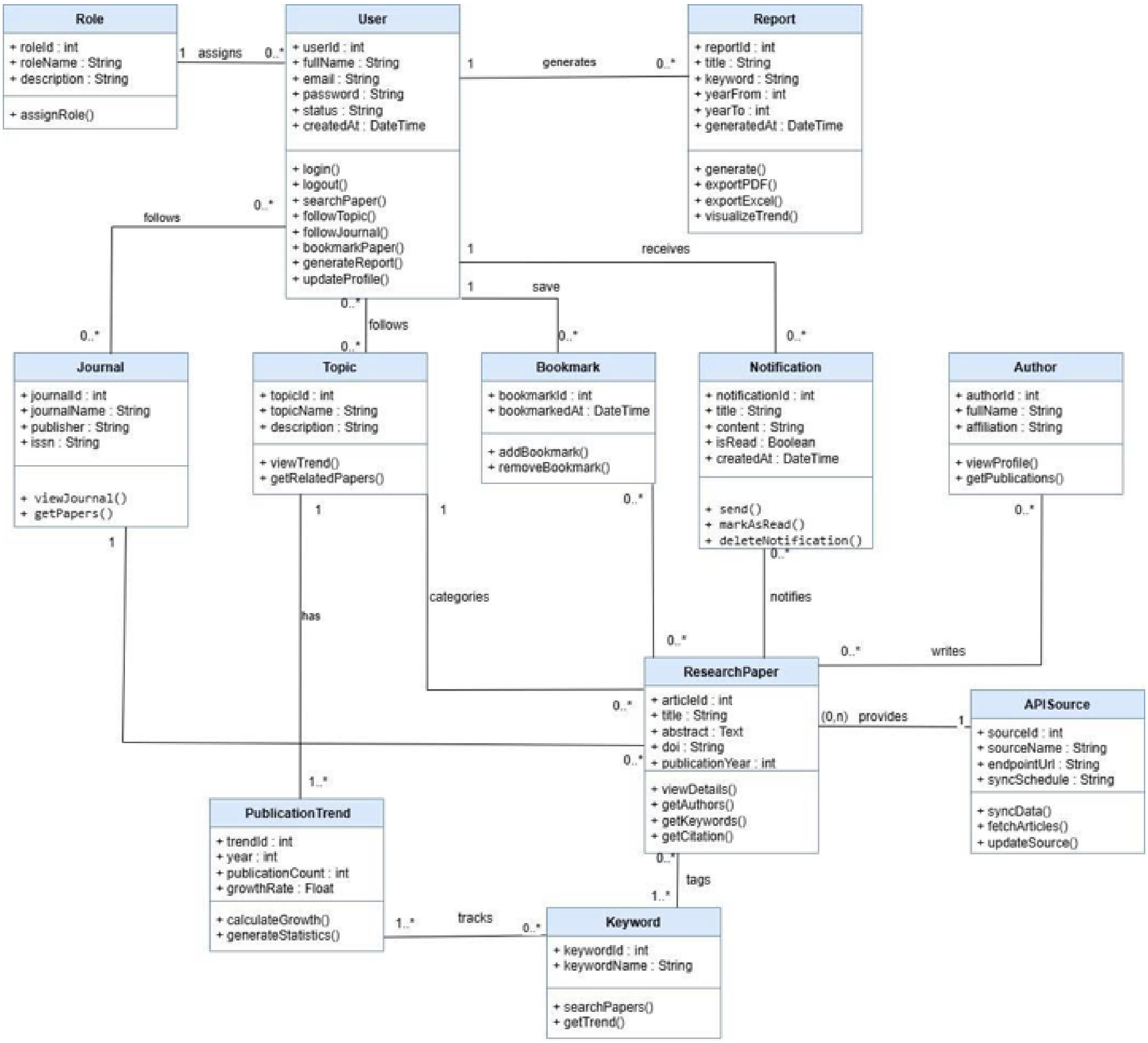
#	Feature	System Function	Description
1	Automated Academic Data Synchronization	Scheduled Job	Scheduled job that automatically retrieves publication metadata from external academic APIs (Semantic Scholar, OpenAlex, Crossref). Inserts new records, updates existing ones, removes duplicates, and logs synchronization statistics in the database.
2	Automated Notification Generation	Scheduled Job	Scheduled job that monitors followed topics for each user. When new publications related to followed topics are detected after synchronization, the system automatically generates and stores notifications in the database for the relevant users.
3	Automated Trend Calculation	Scheduled Job	Scheduled job that recalculates publication trend statistics after each data synchronization. Updates keyword frequency, topic growth rate, and journal distribution data used by the dashboard and trending topics features.

2.7 Entity Relationship Diagram



Hình 1: Entity Relationship Diagram (ERD)

2.8 Class Diagram



Hình 2: Class Diagram

IV. Software Design Description

1 System Design

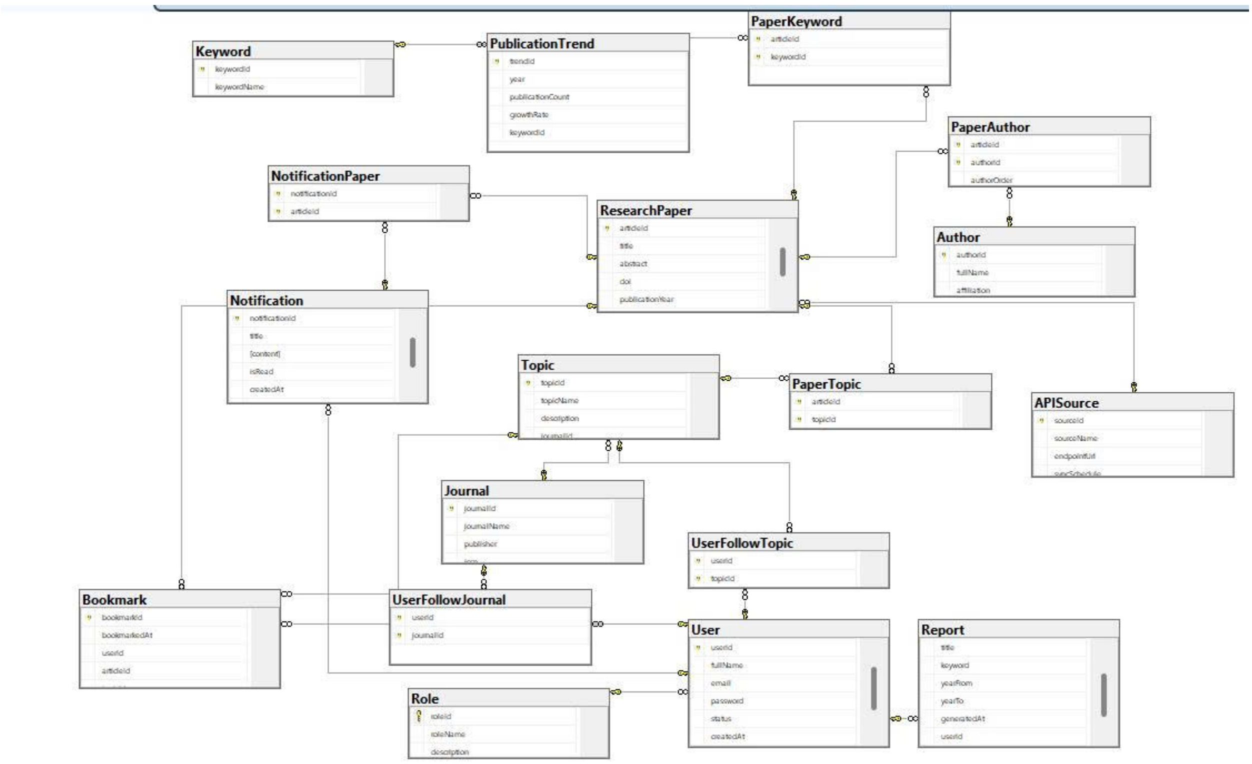
1.1 System Architecture

1.2 Package Diagram

1.2.1 Front End

1.2.2 Back End

2 Database Design



Hình 3: Database Design

<u>**Entities Description**</u>

2.1 Role Table

Field	Type	Description	PK	FK
roleId	INT	Role ID	Yes	No
roleName	NVARCHAR(100)	Role name	No	No
description	NVARCHAR(255)	Description	No	No

2.2 User Table

Field	Type	Description	PK	FK
userId	INT	User ID	Yes	No
fullName	NVARCHAR(150)	Full name	No	No
email	NVARCHAR(150)	Email	No	No
password	NVARCHAR(255)	Password	No	No
status	NVARCHAR(50)	Status	No	No
createdAt	DATETIME2	Created time	No	No
roleId	INT	Role reference	No	Yes → Role

2.3 Journal Table

Field	Type	Description	PK	FK
journalId	INT	Journal ID	Yes	No
journalName	NVARCHAR(200)	Journal name	No	No
publisher	NVARCHAR(150)	Publisher	No	No
issn	NVARCHAR(20)	ISSN	No	No

2.4 UserFollowJournal (M:N)

Field	Type	Description	PK	FK

userId	INT	User	Yes (Composite)	Yes → User
journalId	INT	Journal	Yes (Composite)	Yes → Journal

2.5 Topic Table

Field	Type	Description	PK	FK
topicId	INT	Topic ID	Yes	No
topicName	NVARCHAR(150)	Topic name	No	No
description	NVARCHAR(MAX)	Description	No	No
journalId	INT	Journal reference	No	Yes → Journal

2.6 UserFollowTopic (M:N)

Field	Type	Description	PK	FK
userId	INT	User	Yes (Composite)	Yes → User
topicId	INT	Topic	Yes (Composite)	Yes → Topic

2.7 APISource Table

Field	Type	Description	PK	FK
sourceId	INT	Source ID	Yes	No
sourceName	NVARCHAR(150)	Source name	No	No
endpointUrl	NVARCHAR(500)	API URL	No	No
syncSchedule	NVARCHAR(100)	Sync schedule	No	No

2.8 Author Table

Field	Type	Description	PK	FK
authorId	INT	Author ID	Yes	No
fullName	NVARCHAR(150)	Full name	No	No
affiliation	NVARCHAR(255)	Affiliation	No	No

2.9 Keyword Table

Field	Type	Description	PK	FK
keywordId	INT	Keyword ID	Yes	No
keywordName	NVARCHAR(100)	Keyword	No	No

2.10 Research Paper Table

Field	Type	Description	PK	FK
articleId	INT	Paper ID	Yes	No
title	NVARCHAR(500)	Title	No	No
abstract	NVARCHAR(MAX)	Abstract	No	No
doi	NVARCHAR(200)	DOI	No	No
publicationYear	INT	Year	No	No
sourceId	INT	API Source	No	Yes → APISource

2.11 PaperTopic (M:N)

Field	Type	Description	PK	FK
articleId	INT	Paper	Yes (Composite)	Yes → ResearchPaper

topicId	INT	Topic	Yes (Composite)	Yes → Topic

2.12 PaperKeyword (M:N)

Field	Type	Description	PK	FK
articleId	INT	Paper	Yes (Composite)	Yes → ResearchPaper
keywordId	INT	Keyword	Yes (Composite)	Yes → Keyword

2.13 PaperAuthor (M:N)

Field	Type	Description	PK	FK
articleId	INT	Paper	Yes (Composite)	Yes → ResearchPaper
authorId	INT	Author	Yes (Composite)	Yes → Author
authorOrder	INT	Author order	No	No

2.14 PublicationTrend Table

Field	Type	Description	PK	FK
trendId	INT	Trend ID	Yes	No
year	INT	Year	No	No
publicationCount	INT	Count	No	No
growthRate	FLOAT	Growth rate	No	No

keywordId	INT	Keyword	No	Yes → Keyword

2.15 Report Table

Field	Type	Description	PK	FK
reportId	INT	Report ID	Yes	No
title	NVARCHAR(300)	Title	No	No
keyword	NVARCHAR(200)	Keyword	No	No
yearFrom	INT	From year	No	No
yearTo	INT	To year	No	No
generatedAt	DATETIME2	Created time	No	No
userId	INT	User	No	Yes → User

2.16 Notification Table

Field	Type	Description	PK	FK
notificationId	INT	Notification ID	Yes	No
title	NVARCHAR(300)	Title	No	No
content	NVARCHAR(MAX)	Content	No	No
isRead	BIT	Read status	No	No
createdAt	DATETIME2	Created time	No	No
userId	INT	User	No	Yes → User

2.17 NotificationPaper (M:N)

Field	Type	Description	PK	FK

notificationId	INT	Notification	Yes (Composite)	Yes → Notification
articleId	INT	Paper	Yes (Composite)	Yes → ResearchPaper

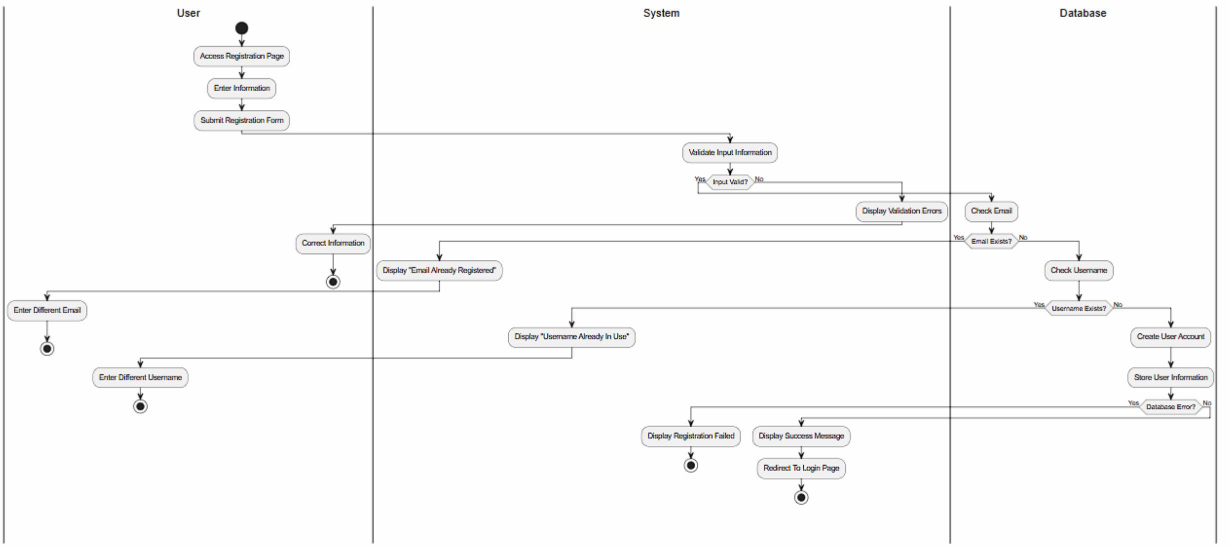
2.18 Bookmark Table

Field	Type	Description	PK	FK
bookmarkId	INT	Bookmark ID	Yes	No
bookmarkedAt	DATETIME2	Time saved	No	No
userId	INT	User	No	Yes → User
articleId	INT	Paper	No	Yes → ResearchPaper
topicId	INT	Topic	No	Yes → Topic

3 Detail Design

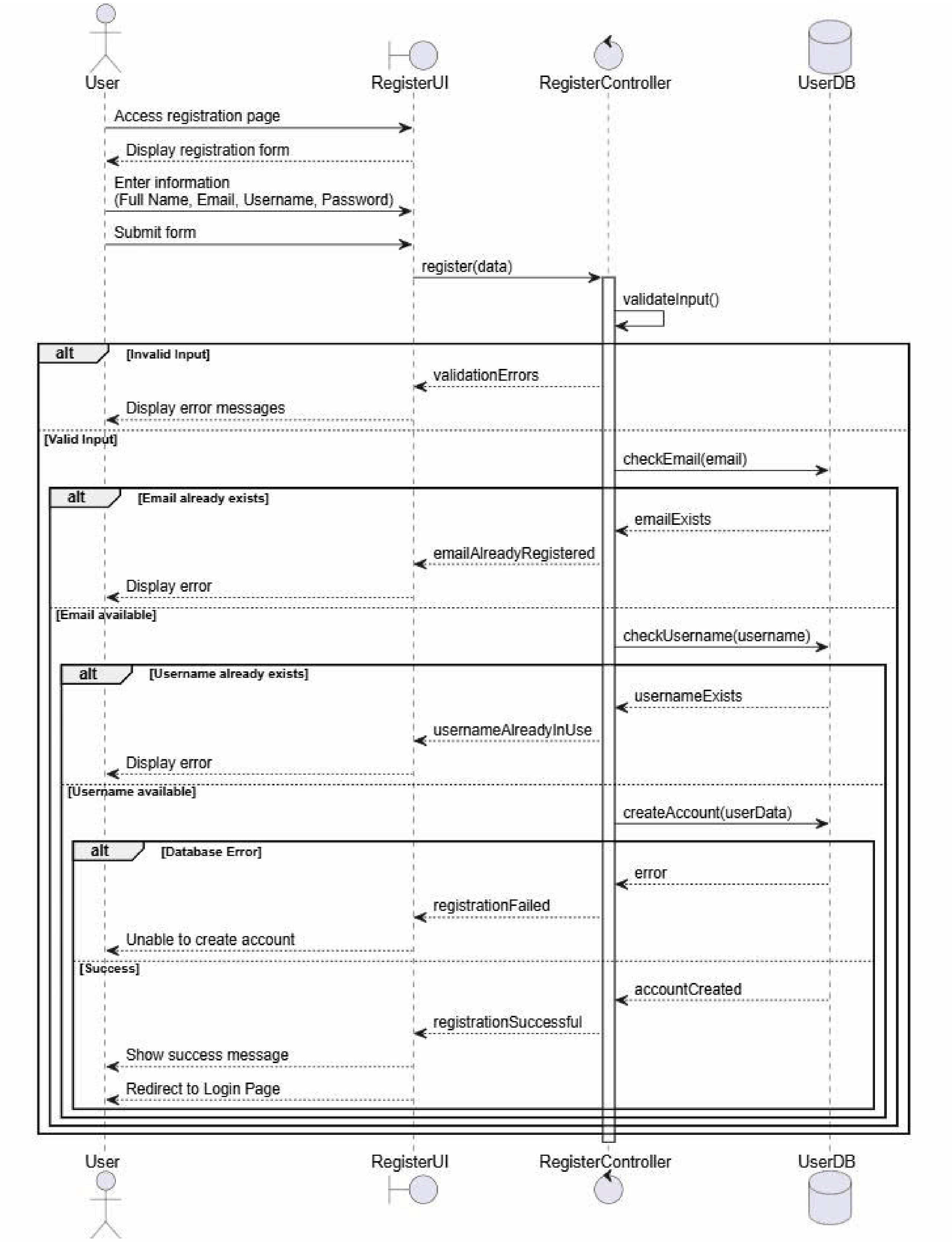
3.1 Register

3.1.1 Activity diagram



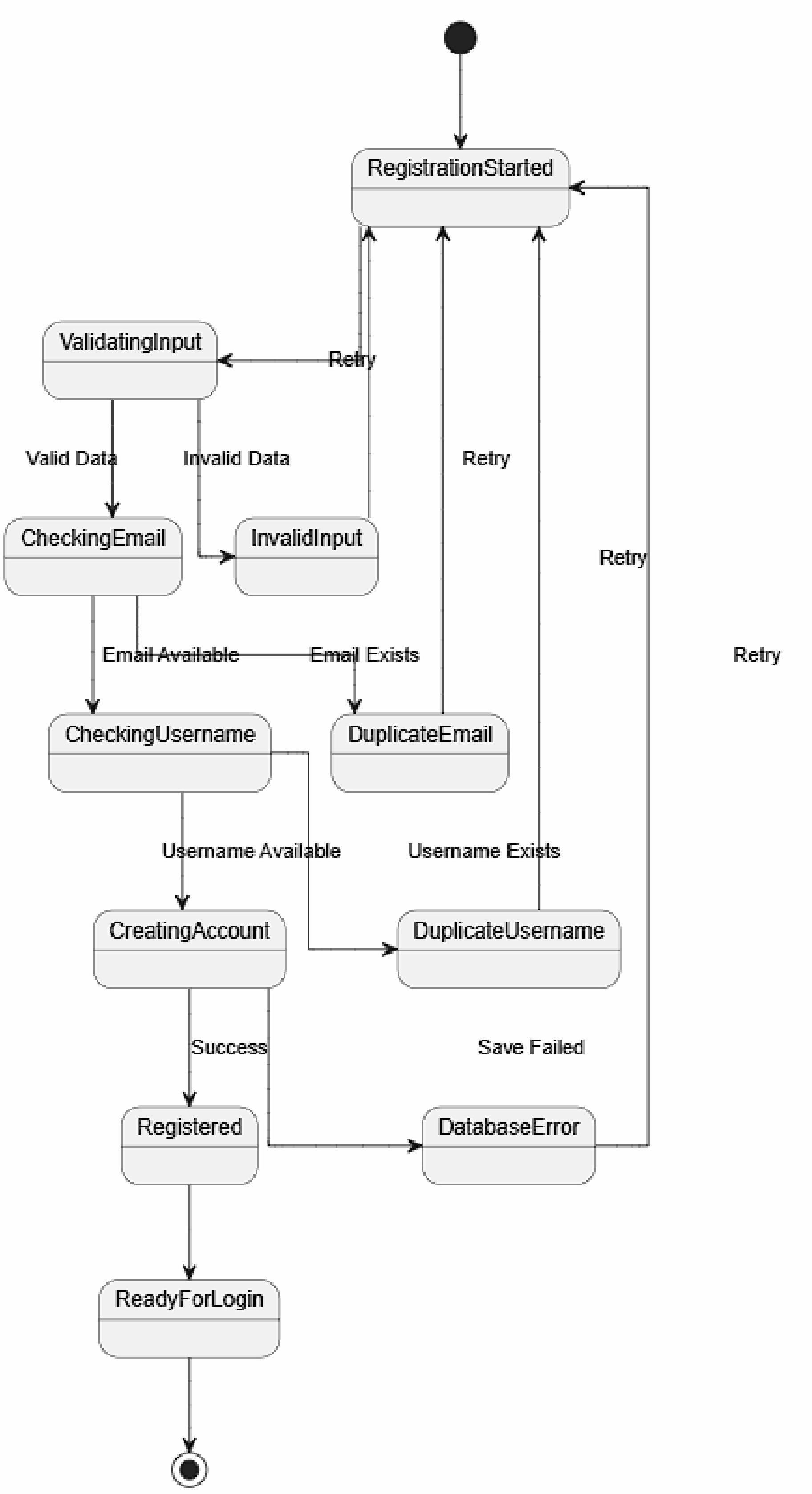
Hình 4: Activity diagram

3.1.2 Sequence Diagram



Hình 5: Sequence Diagram

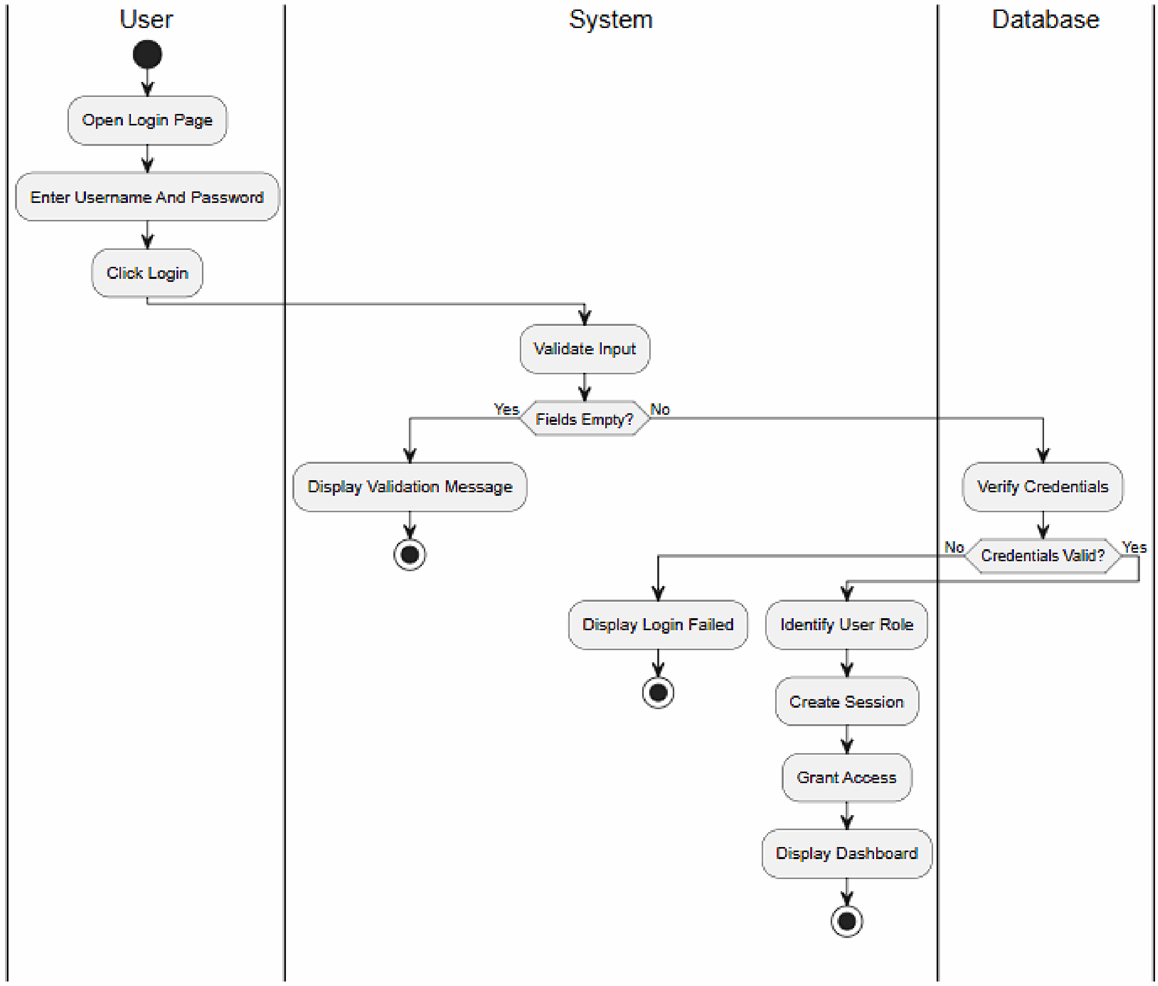
3.1.3 State Diagram



Hình 6: State diagram

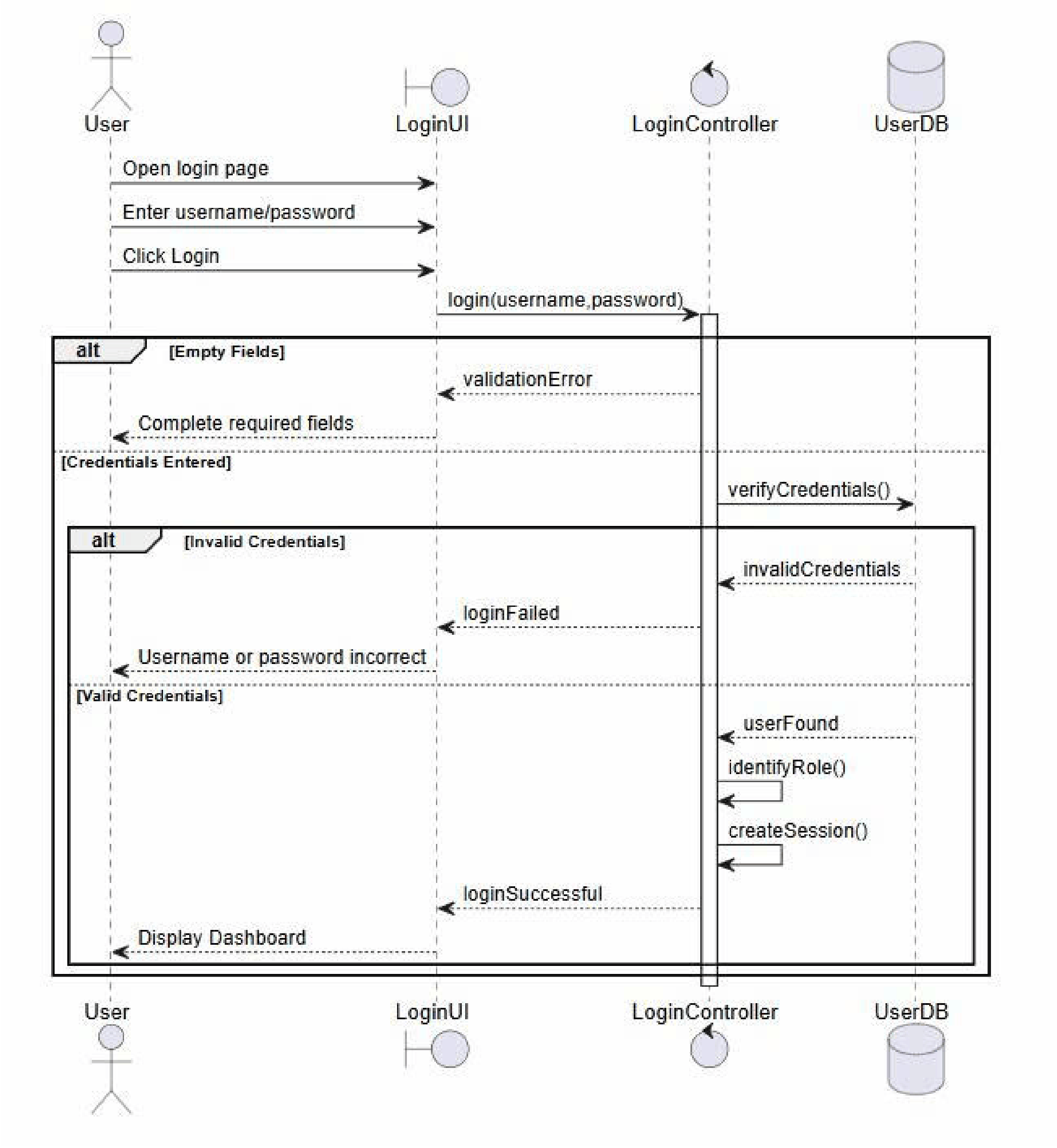
3.2 Login

3.2.1 Activity Diagram



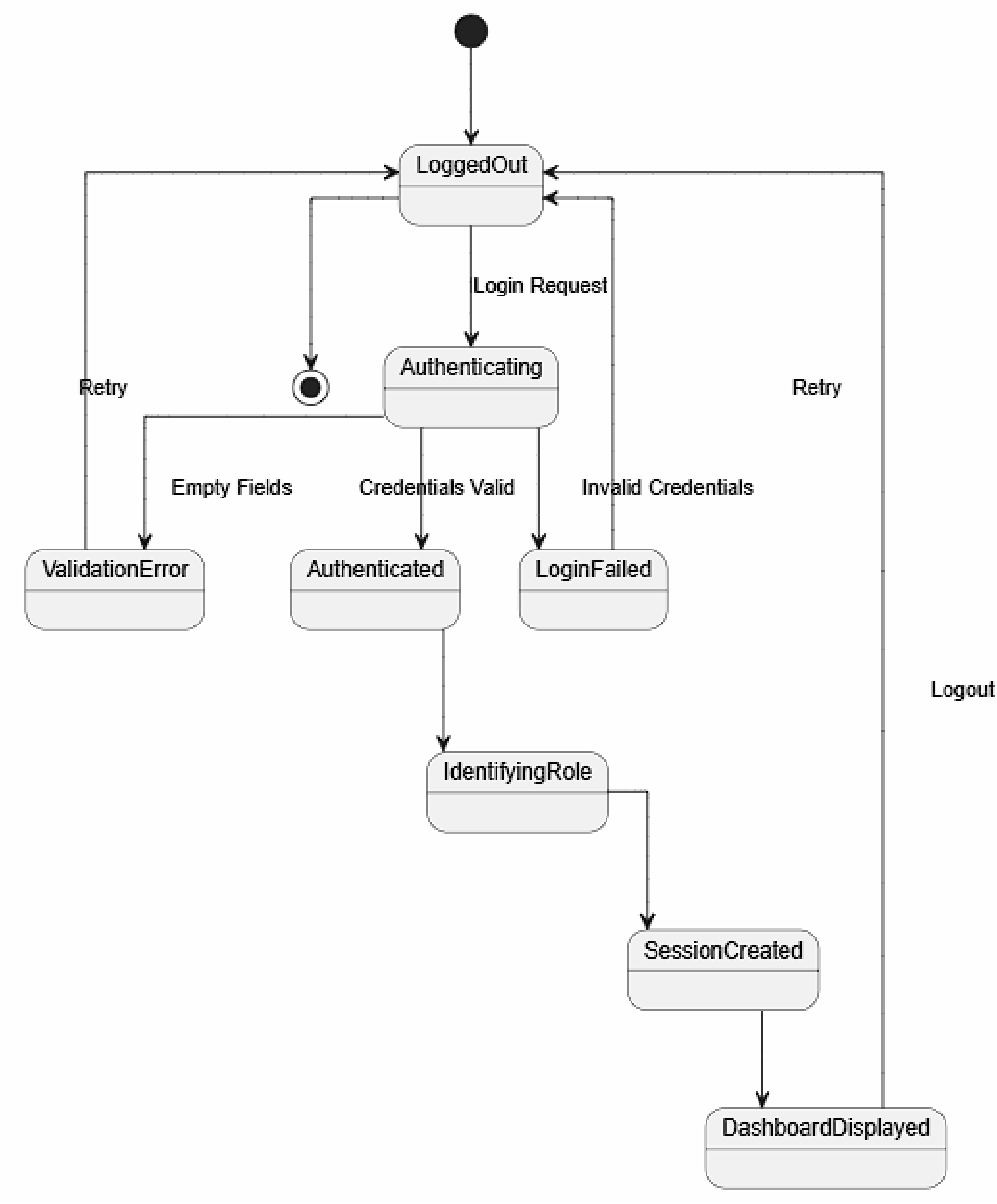
Hình 7: Activity diagram

3.2.2 Sequence Diagram



Hình 8: Sequence Diagram

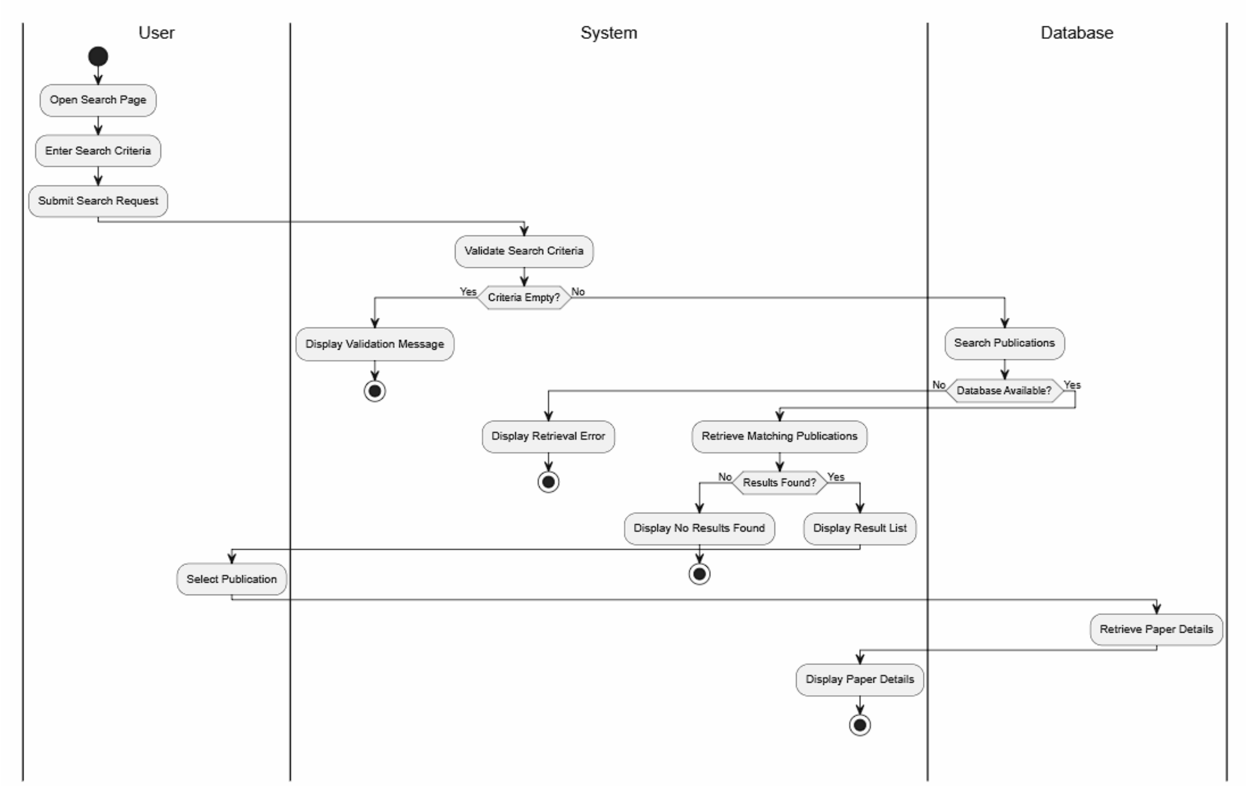
3.2.3 State Diagram



Hình 9: State diagram

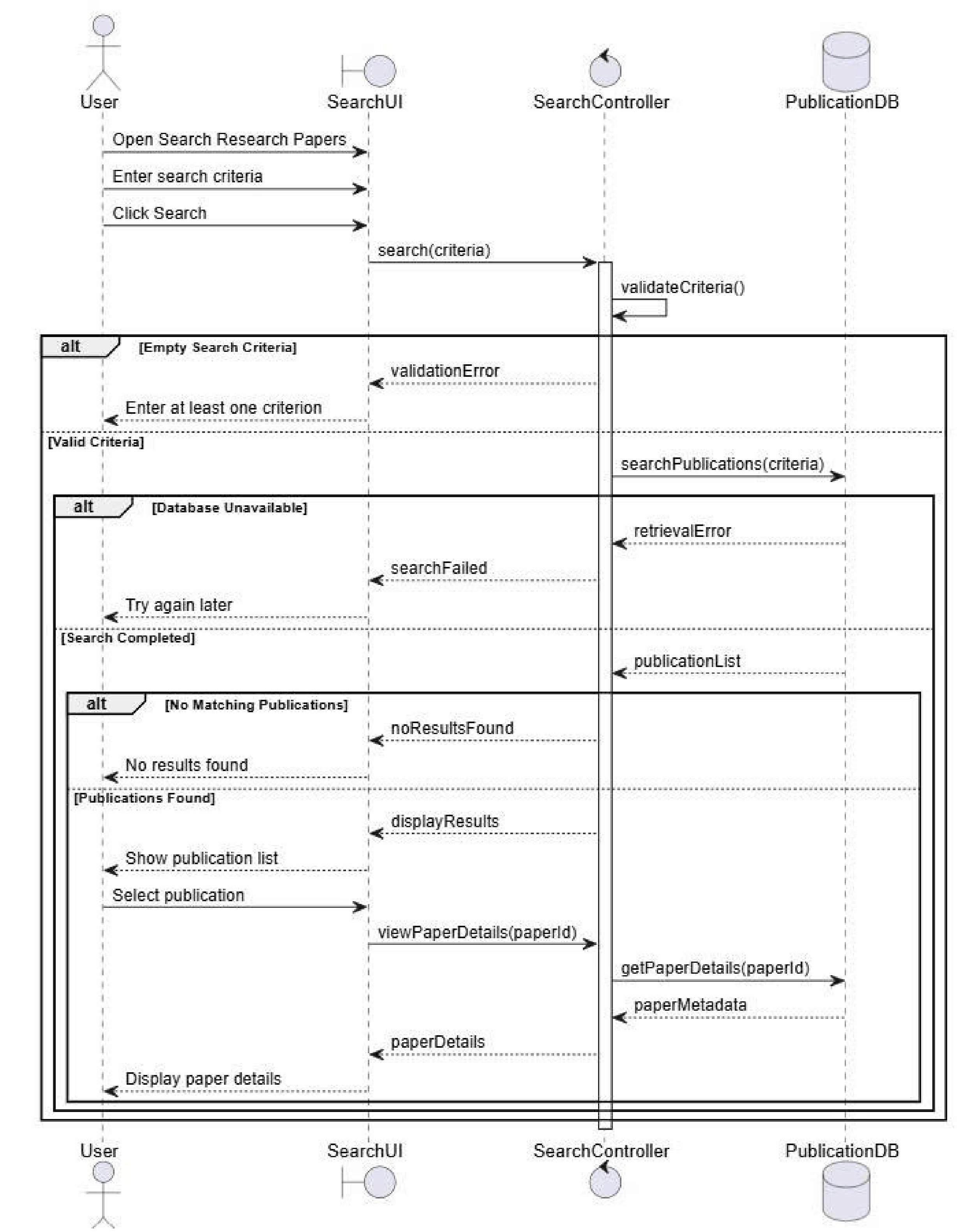
3.3 Search Research Papers

3.3.1 Activity Diagram



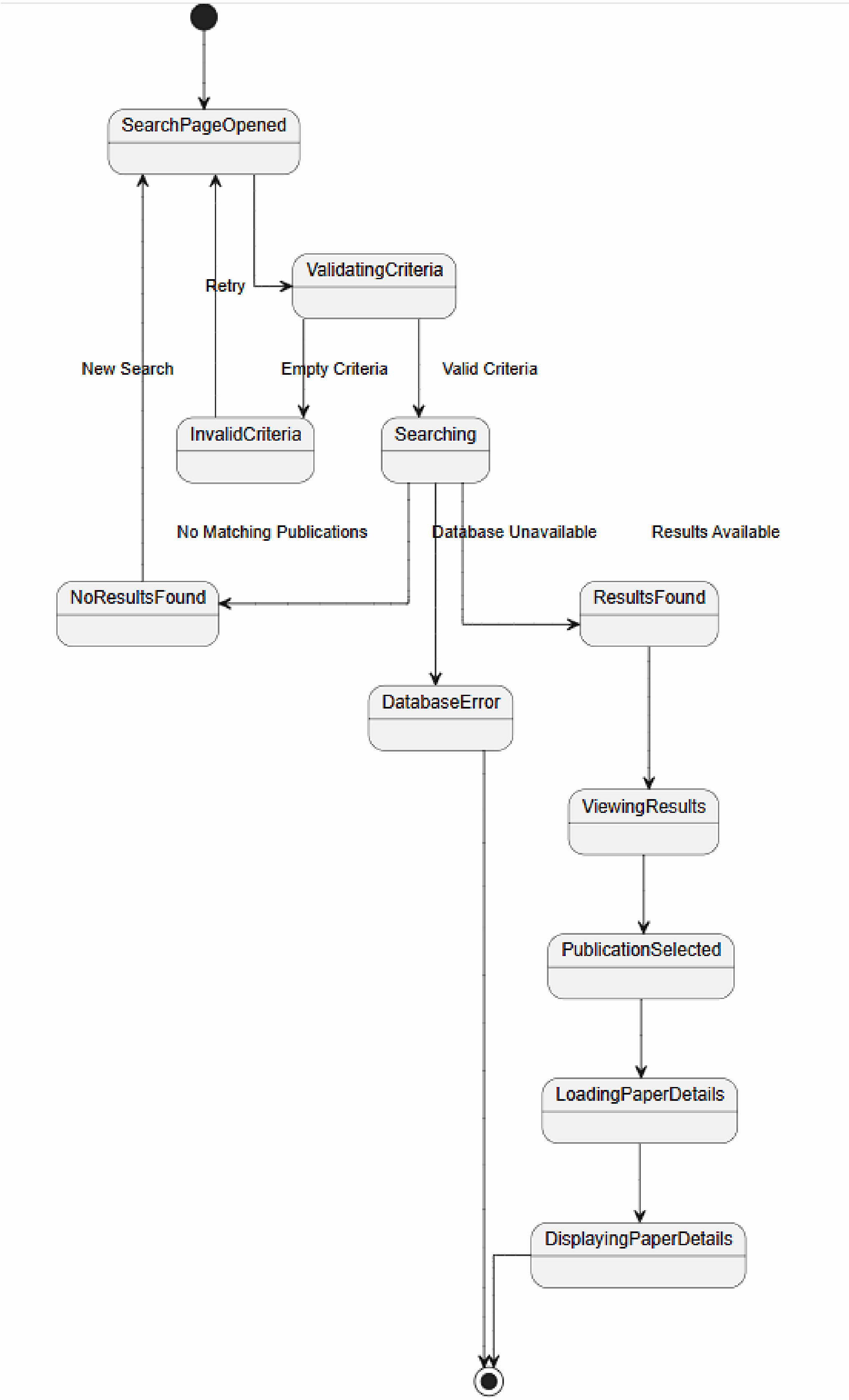
Hình 10: Activity diagram

3.3.2 Sequence Diagram



Hình 11: Sequence Diagram

3.3.3 State Diagram



Hình 12: State diagram

3.4 View Paper Details

3.4.1 Activity Diagram

3.4.2 Sequence Diagram

3.4.3 State Diagram

3.5 Save Bookmarks

3.5.1 Activity Diagram

3.5.2 Sequence Diagram

3.5.3 State Diagram

3.6 Track Publication Trends

3.6.1 Activity Diagram

3.6.2 Sequence Diagram

3.6.3 State Diagram

3.7 View Dashboard

3.7.1 Activity Diagram

3.7.2 Sequence Diagram

3.7.3 State Diagram

3.8 Generate Reports

3.8.1 Activity Diagram

3.8.2 Sequence Diagram

3.8.3 State Diagram

3.9 View Trending Topics

3.9.1 Activity Diagram

3.9.2 Sequence Diagram

3.9.3 State Diagram

3.10 Follow Topics

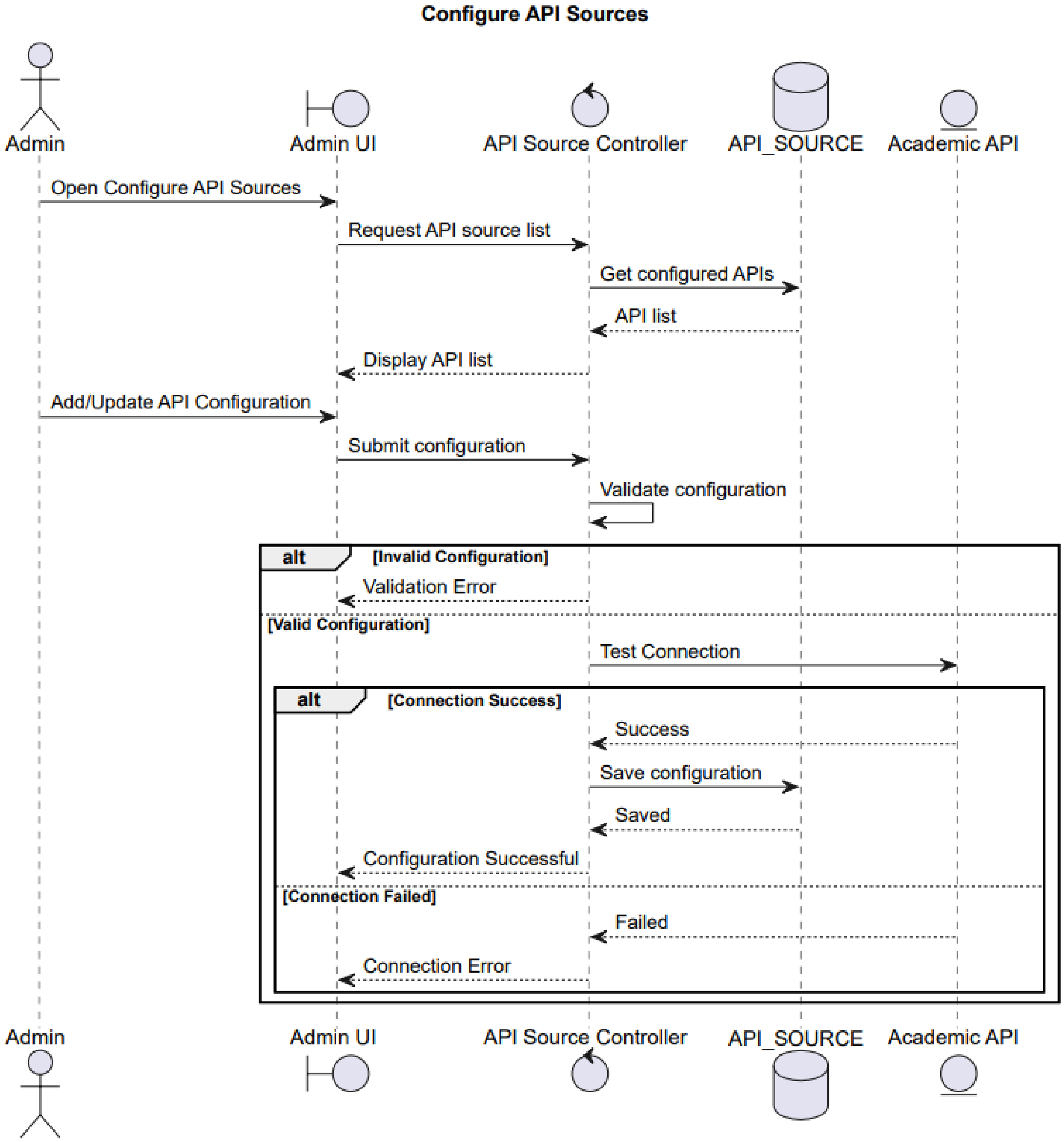
3.10.1 Activity Diagram

3.10.2 Sequence Diagram

3.10.3 State Diagram

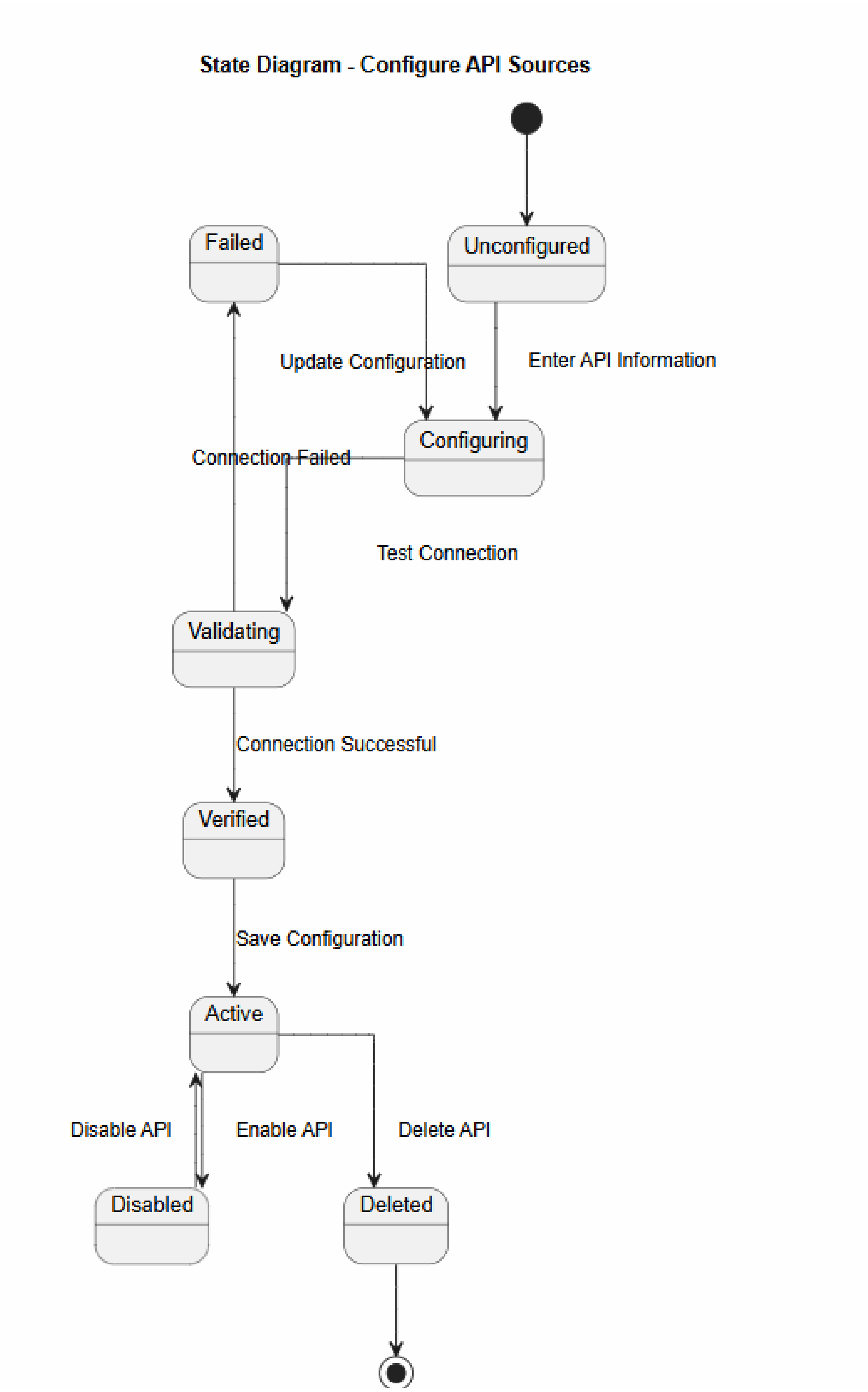
3.11 Receive Notifications

3.14.2 Sequence Diagram



Hình 14: Configure API Sources

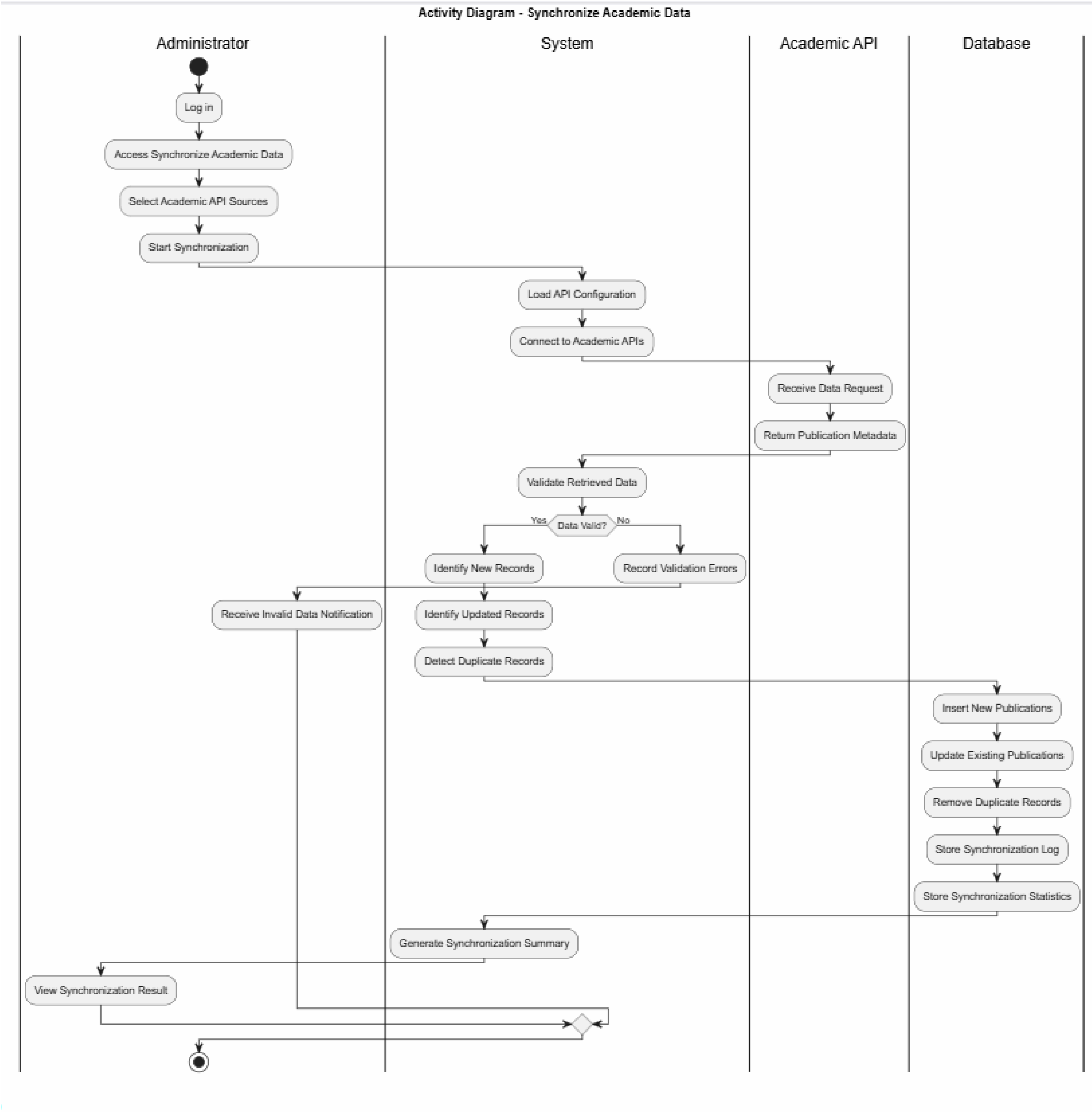
3.14.3 State Diagram



Hình 15: State diagram

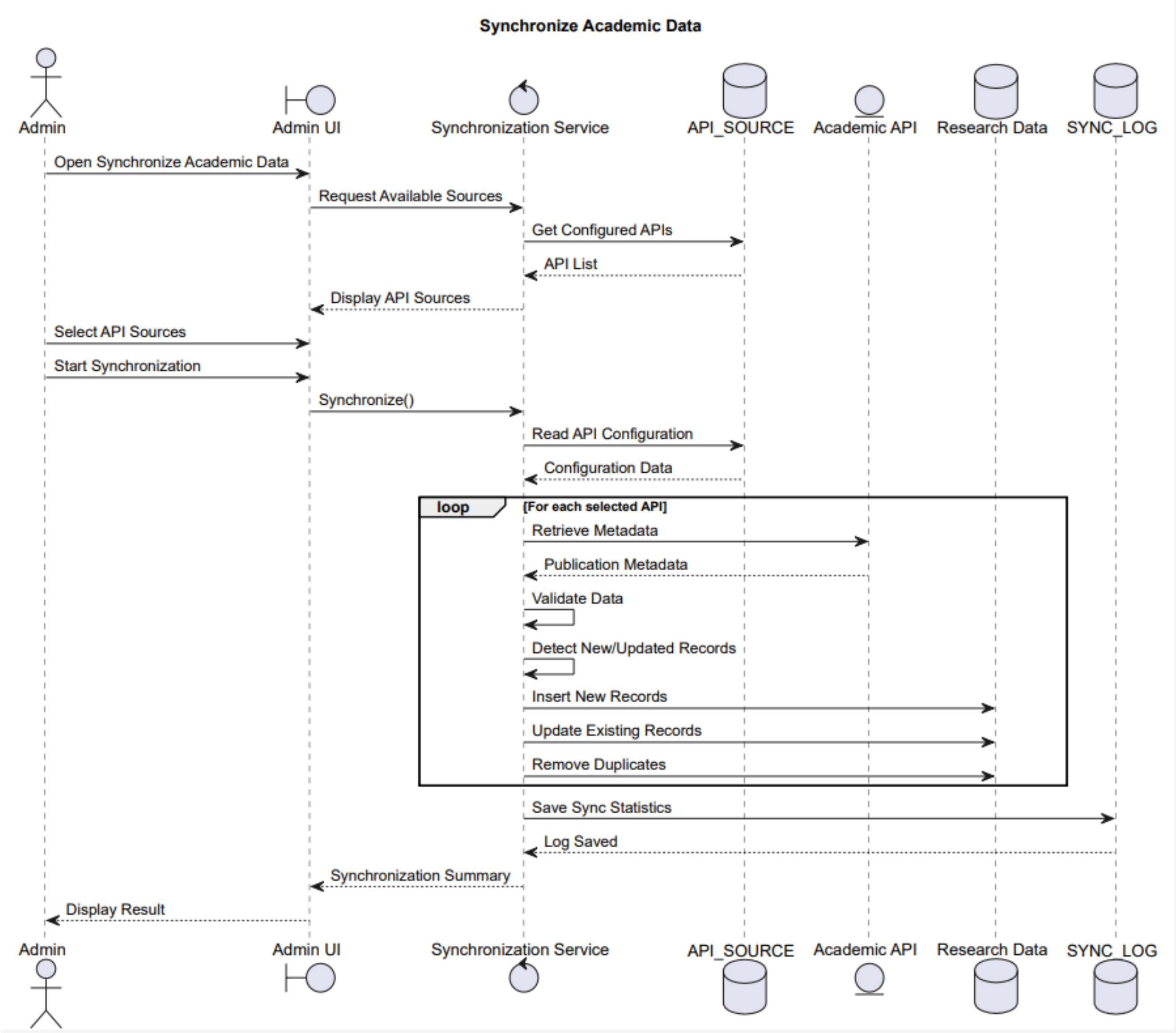
3.15 Sync Academic Data

3.15.1 Activity Diagram



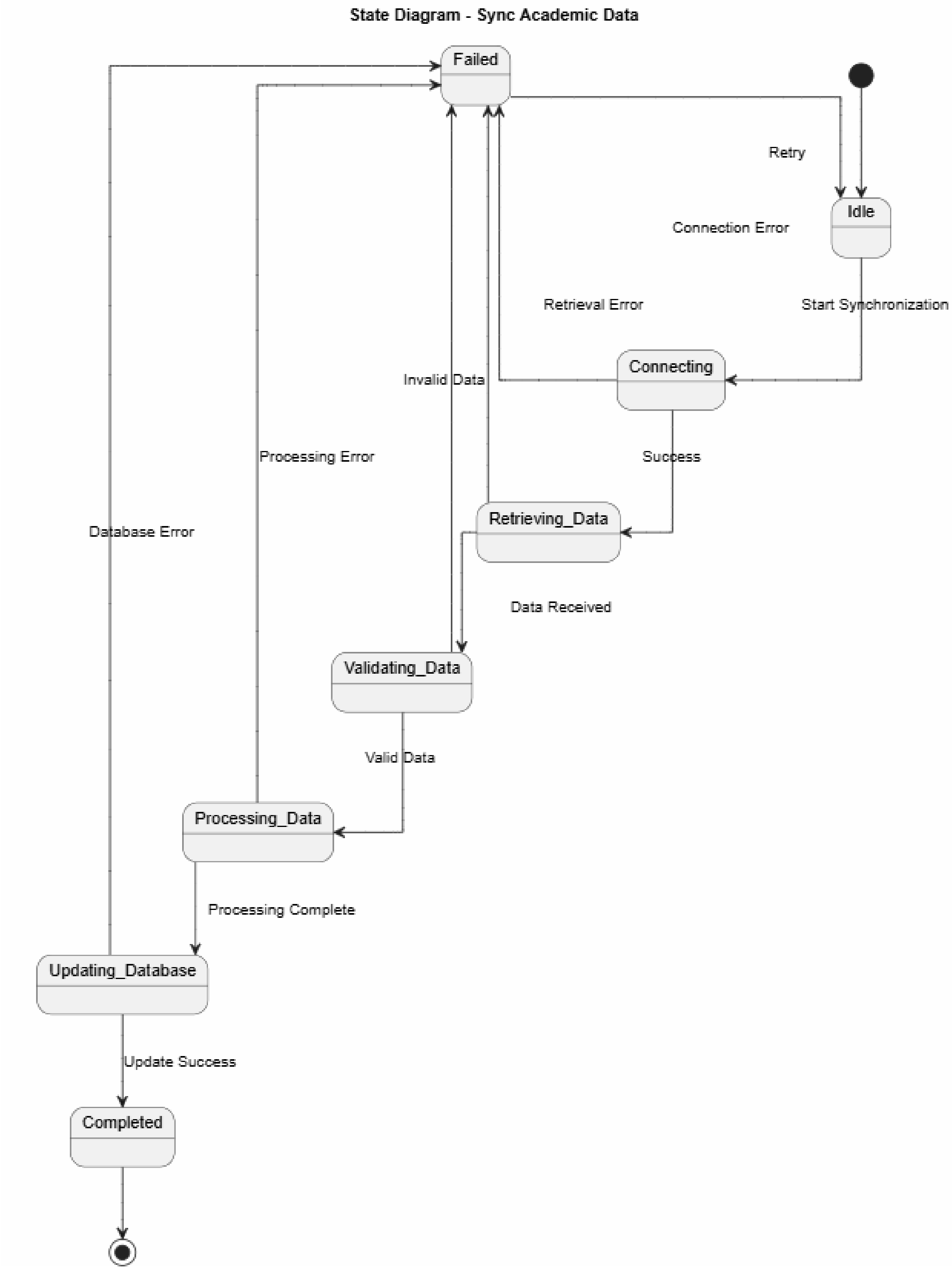
Hình 16: Sync Academic Data

3.15.2 Sequence Diagram



Hình 17: Sync Academic Data

3.15.3 State Diagram



Hình 18: Sync Academic Data